

HUMANITY-ONE MATERIAL SOLUTIONS

Complete technical-operational report for deposit, institutional evaluation and authorized use

Author: Pierernesto Di Paolo

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Authorization requests: pierernesto.di.paolo.69@proton.me

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Operational notice

This document does not present a validated clinical cure, does not replace physicians, laboratories, institutions or competent authorities, and does not provide wet-lab procedures, genetic editing, cellular manipulation, drug prescription or unauthorized biological experimentation. Its function is to convert measured human risks into material pathways for resolution, access, funding, public responsibility, referral to authorized structures and outcome evaluation.

This version incorporates into the body of the document the operational registers that make the system executable: problem catalog, cost table, role map, physical inventory, pathway library, pre-filled Material Solution Units, demonstrative cases, Real Resolution Score, Cost of Inaction, public dossiers, funding request, laboratory handoff and legal collapse impact.

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1. System purpose

HUMANITY-ONE Material Solutions is designed to reduce the gap between problem identification and the production of a material solution. The system starts from real risks, such as severe pain, missing medication, economic collapse, denied access, isolation, functional decline or inaccessible advanced therapy, and converts them into operational units with trigger, required material, authorized actor, estimated cost, funding channel, access pathway, fallback and outcome metric.

The system does not claim that every problem is already solved. It states that no detected problem should remain without classification, pathway, responsible actor, cost, fallback, dossier or measurable outcome.

2. Material architecture

The operational architecture is based on seven factories: Physical Care Factory, Economic Security Factory, Mental Happiness Factory, Longevity Prevention Factory, Advanced Therapy Access Factory, EDEN/Lab Invention Factory, and Legal and State Responsibility Factory. Each factory produces material outputs, not only descriptions: requests, referrals, registers, dossiers, funds, laboratory handoff, institutional escalation or measured outcomes.

The success criterion is not the existence of a document, but the presence of at least one of these outputs: SOLVE_NOW, CLINICAL_ROUTE, ECONOMIC_RESCUE, PUBLIC_DUTY, FUNDING_ROUTE, LAB_ROUTE, TRIAL_ROUTE, INVENTION_ROUTE, LEGAL_ROUTE or UNRESOLVED_MONITOR.

3. Canonical register 1 - Problem Catalog

The following catalog is the master table of problems. Each entry defines the problem, trigger, severity, urgency, target factory and pathway to activate.

Problem P001 - PAIN_SEVERE

Domain: PAIN/CLINICAL

Trigger: NRS ≥ 7

Severity 0-5: 5

Urgency 0-5: 5

Default root cause: Untreated pain / access barrier / clinical cause

Target factory: Physical Care Factory

MSU required: YES

Primary pathway: PATH_PAIN_48H

Fallback 1: Clinical referral escalation

Fallback 2: Public responsibility if access blocked

Required output: CLINICAL_ROUTE

Problem P002 - MEDICATION_MISSING

Domain: ACCESS/ECON

Trigger: Essential medication unavailable

Severity 0-5: 5

Urgency 0-5: 5

Default root cause: Cost, prescription, stock, bureaucracy

Target factory: Physical Care + Economic Security

MSU required: YES

Primary pathway: PATH_MEDICATION_ACCESS

Fallback 1: Economic Rescue

Fallback 2: Public Responsibility

Required output: SOLVE_NOW

Problem P003 - HOUSING_CRITICAL

Domain: ECON/SOCIAL

Trigger: EFS_2 <= 1

Severity 0-5: 5

Urgency 0-5: 5

Default root cause: Housing instability / eviction / unsafe housing

Target factory: Economic Security Factory

MSU required: YES

Primary pathway: PATH_HOUSING_RESCUE

Fallback 1: Municipal emergency route

Fallback 2: Public Responsibility

Required output: ECONOMIC_RESCUE

Problem P004 - FOOD_CRITICAL

Domain: ECON

Trigger: EFS_3 <= 1

Severity 0-5: 5

Urgency 0-5: 5

Default root cause: Food insecurity / no liquidity

Target factory: Economic Security Factory

MSU required: YES

Primary pathway: PATH_FOOD_SECURITY

Fallback 1: Foundation/charity bridge

Fallback 2: Public Responsibility

Required output: ECONOMIC_RESCUE

Problem P005 - CARE_ACCESS_BLOCKED

Domain: ACCESS

Trigger: A_raw >= 4 or missed critical visit

Severity 0-5: 4

Urgency 0-5: 4

Default root cause: Service unavailability / waitlist / bureaucracy

Target factory: Physical Care Factory

MSU required: YES

Primary pathway: PATH_CLINICAL_REFERRAL

Fallback 1: Alternative center

Fallback 2: Public Responsibility

Required output: CLINICAL_ROUTE

Problem P006 - DEPRESSION_HIGH

Domain: MENTAL

Trigger: PHQ-9 >= 15 or item 9 positive

Severity 0-5: 5

Urgency 0-5: 5

Default root cause: Depression, pain, poverty, isolation

Target factory: Mental Happiness Factory

MSU required: YES

Primary pathway: PATH_MENTAL_HEALTH

Fallback 1: CSM/urgent clinical route

Fallback 2: Safety escalation

Required output: CLINICAL_ROUTE

[Problem P007 - ANXIETY_HIGH](#)

Domain: MENTAL

Trigger: GAD-7 ≥ 15

Severity 0-5: 4

Urgency 0-5: 3

Default root cause: Anxiety, uncertainty, social/economic stress

Target factory: Mental Happiness Factory

MSU required: YES

Primary pathway: PATH_ANXIETY_SUPPORT

Fallback 1: Primary care referral

Fallback 2: Psychological support

Required output: CLINICAL_ROUTE

[Problem P008 - ISOLATION_HIGH](#)

Domain: SOCIAL

Trigger: UCLA high / social support ≤ 1

Severity 0-5: 4

Urgency 0-5: 3

Default root cause: Loneliness / mobility / stigma / poverty

Target factory: Mental Happiness + Social Factory

MSU required: YES

Primary pathway: PATH_ANTI_ISOLATION

Fallback 1: Community partner

Fallback 2: Public Responsibility

Required output: SOLVE_NOW

[Problem P009 - ECONOMIC_COLLAPSE](#)

Domain: ECON

Trigger: EFS <= 14

Severity 0-5: 5

Urgency 0-5: 5

Default root cause: Income/housing/food/health-cost collapse

Target factory: Economic Security Factory

MSU required: YES

Primary pathway: PATH_ECONOMIC_RESCUE

Fallback 1: Foundation bridge

Fallback 2: Public Responsibility

Required output: ECONOMIC_RESCUE

Problem P010 - AGING_RISK_HIGH

Domain: LONGEVITY

Trigger: Functional decline or biomarkers abnormal

Severity 0-5: 3

Urgency 0-5: 2

Default root cause: Metabolic/inflammatory/functional aging drivers

Target factory: Longevity Prevention Factory

MSU required: YES

Primary pathway: PATH_LONGEVITY_BASELINE

Fallback 1: Clinical preventive route

Fallback 2: Research route

Required output: CLINICAL_ROUTE

Problem P011 - ADVANCED_THERAPY_INACCESSIBLE

Domain: ACCESS/ADVANCED

Trigger: Approved therapy exists but not accessible

Severity 0-5: 5

Urgency 0-5: 4

Default root cause: Cost, eligibility, center, coverage barrier

Target factory: Advanced Therapy Access Factory

MSU required: YES

Primary pathway: PATH_ADVANCED_THERAPY_ACCESS

Fallback 1: Funding route

Fallback 2: Public Responsibility

Required output: FUNDING_ROUTE

Problem P012 - GENETIC_DISEASE_TARGET

Domain: GENETIC

Trigger: Diagnosis may match approved/trial therapy

Severity 0-5: 5

Urgency 0-5: 4

Default root cause: Monogenic/rare disease target

Target factory: Advanced Therapy Access Factory

MSU required: YES

Primary pathway: PATH_GENE_CELL_ACCESS

Fallback 1: Trial route

Fallback 2: Lab/clinical specialist

Required output: TRIAL_ROUTE

Problem P013 - REGENERATION_NEED

Domain: REGEN

Trigger: Function loss with potential recovery/trial

Severity 0-5: 4

Urgency 0-5: 3

Default root cause: Tissue/organ functional loss

Target factory: EDEN/Lab + Regeneration Factory

MSU required: YES

Primary pathway: PATH_REGENERATION_TRIAGE

Fallback 1: Rehabilitation route

Fallback 2: Research route

Required output: LAB_ROUTE

[Problem P014 - UNRESOLVED_BIOLOGY](#)

Domain: RESEARCH

Trigger: No adequate solution exists

Severity 0-5: 4

Urgency 0-5: 2

Default root cause: Scientific/technology gap

Target factory: EDEN/Lab Invention Factory

MSU required: YES

Primary pathway: PATH_EDEN_LAB_HANDOFF

Fallback 1: Grant concept

Fallback 2: Invention backlog

Required output: INVENTION_ROUTE

[Problem P015 - LEGAL_ACCESS_HARM](#)

Domain: LEGAL

Trigger: Law/procedure blocks health/economy/access

Severity 0-5: 4

Urgency 0-5: 3

Default root cause: Bureaucratic/legal barrier

Target factory: Legal & State Responsibility Factory

MSU required: YES

Primary pathway: PATH_LEGAL_COLLAPSE_REVIEW

Fallback 1: Escalation dossier

Fallback 2: Public Responsibility

Required output: LEGAL_ROUTE

[Problem P016 - FUNDING_GAP](#)

Domain: FUNDING

Trigger: Solution exists but payer absent

Severity 0-5: 5

Urgency 0-5: 4

Default root cause: Affordability/funding barrier

Target factory: Funding Engine

MSU required: YES

Primary pathway: PATH_FUNDING_REQUEST

Fallback 1: Public duty dossier

Fallback 2: Foundation route

Required output: FUNDING_ROUTE

Problem P017 - TRANSPORT_BARRIER

Domain: ACCESS/ECON

Trigger: Cannot physically reach service

Severity 0-5: 3

Urgency 0-5: 3

Default root cause: Transport cost/distance/disability

Target factory: Economic + Access Factory

MSU required: YES

Primary pathway: PATH_TRANSPORT_ACCESS

Fallback 1: Local service alternative

Fallback 2: Public Responsibility

Required output: SOLVE_NOW

Problem P018 - DIGITAL_BARRIER

Domain: ACCESS

Trigger: Cannot use portals/forms/digital access

Severity 0-5: 3

Urgency 0-5: 2

Default root cause: Digital exclusion

Target factory: Access Factory

MSU required: YES

Primary pathway: PATH_DIGITAL_SUPPORT

Fallback 1: Paper route

Fallback 2: Social service support

Required output: SOLVE_NOW

[Problem P019 - DEBT_HEALTH_BLOCK](#)

Domain: ECON

Trigger: EFS_6 <= 1

Severity 0-5: 4

Urgency 0-5: 4

Default root cause: Debt or health cost blocks essentials

Target factory: Economic Security Factory

MSU required: YES

Primary pathway: PATH_DEBT_HEALTH_STABILIZATION

Fallback 1: Legal aid/consumer support

Fallback 2: Public Responsibility

Required output: ECONOMIC_RESCUE

[Problem P020 - LOW_MEANING_FUTURE](#)

Domain: MENTAL/IDENTITY

Trigger: MLQ low or LOT-R low

Severity 0-5: 3

Urgency 0-5: 2

Default root cause: Loss of meaning/future orientation

Target factory: Mental Happiness Factory

MSU required: YES

Primary pathway: PATH_MEANING_FUTURE

Fallback 1: Social connection

Fallback 2: Economic/health root cause

Required output: SOLVE_NOW

4. Canonical register 2 - Cost Table

Costs are indicative euro ranges and must be locally validated. Their purpose is not to define universal prices, but to prevent activation of a pathway without at least an order of magnitude, public/private channel and fallback.

Cost C001 - Essential medication bridge

Category: medication

Cost level: C1-C3

Min EUR: 5

Max EUR: 150

Public cost possible: YES

Private cost possible: YES

Pathway: PATH_MEDICATION_ACCESS

Fallback cost note: If not affordable trigger EFS/Public duty

Local validation required: YES

Notes: Depends on drug and jurisdiction

Cost C002 - Primary care/GP contact

Category: clinical

Cost level: C0-C1

Min EUR: 0

Max EUR: 80

Public cost possible: YES

Private cost possible: YES

Pathway: PATH_CLINICAL_REFERRAL

Fallback cost note: Use public system first

Local validation required: YES

Notes: Costs vary

Cost C003 - Specialist visit

Category: clinical

Cost level: C1-C3

Min EUR: 0

Max EUR: 250

Public cost possible: YES

Private cost possible: YES

Pathway: PATH_CLINICAL_REFERRAL

Fallback cost note: Alternative center if waitlist blocks

Local validation required: YES

Notes: Public/private differ

[Cost C004 - Pain specialist visit](#)

Category: clinical

Cost level: C1-C3

Min EUR: 0

Max EUR: 300

Public cost possible: YES

Private cost possible: YES

Pathway: PATH_PAIN_48H

Fallback cost note: Public duty if severe pain untreated

Local validation required: YES

Notes: Severe pain NRS $\geq$ 7

[Cost C005 - Basic blood panel](#)

Category: laboratory

Cost level: C1-C2

Min EUR: 20

Max EUR: 150

Public cost possible: YES

Private cost possible: YES

Pathway: PATH_LONGEVITY_BASELINE

Fallback cost note: Use essential panel first

Local validation required: YES

Notes: Clinical lab authorized

[Cost C006 - HbA1c/lipids/metabolic panel](#)

Category: laboratory

Cost level: C1-C2

Min EUR: 20

Max EUR: 120

Public cost possible: YES

Private cost possible: YES

Pathway: PATH_LONGEVITY_BASELINE

Fallback cost note: Use public indications where applicable

Local validation required: YES

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

[Cost C007 - hs-CRP/PCR inflammation test](#)

Category: laboratory

Cost level: C1-C2

Min EUR: 10

Max EUR: 80

Public cost possible: YES

Private cost possible: YES

Pathway: PATH_LONGEVITY_BASELINE

Fallback cost note: Optional if not available

Local validation required: YES

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

[Cost C008 - Biological age clock panel](#)

Category: advanced lab

Cost level: C3-C4

Min EUR: 200

Max EUR: 800

Public cost possible: NO/rare

Private cost possible: YES

Pathway: PATH_LONGEVITY_ADVANCED

Fallback cost note: Not required for L0 outcome

Local validation required: YES

Notes: Optional; not primary care standard

Cost C009 - Genetic test targeted

Category: advanced lab

Cost level: C3-C5

Min EUR: 200

Max EUR: 2500

Public cost possible: YES if indicated

Private cost possible: YES

Pathway: PATH_GENE_CELL_ACCESS

Fallback cost note: Only if clinically indicated

Local validation required: YES

Notes: Authorized lab required

Cost C010 - CAR-T/gene/cell therapy evaluation

Category: advanced therapy

Cost level: C4-C5

Min EUR: 0

Max EUR: 10000

Public cost possible: YES case-dependent

Private cost possible: YES

Pathway: PATH_ADVANCED_THERAPY_ACCESS

Fallback cost note: Therapy costs separate; center-based

Local validation required: YES

Notes: Referral/access mapping only

[Cost C011 - Food emergency package](#)

Category: economic/social

Cost level: C1-C2

Min EUR: 20

Max EUR: 300

Public cost possible: YES

Private cost possible: YES

Pathway: PATH_FOOD_SECURITY

Fallback cost note: Food bank/municipal route

Local validation required: YES

Notes: Monthly/local

[Cost C012 - Housing emergency support](#)

Category: economic/social

Cost level: C3-C5

Min EUR: 300

Max EUR: 3000

Public cost possible: YES

Private cost possible: YES

Pathway: PATH_HOUSING_RESCUE

Fallback cost note: Escalate immediately if unsafe

Local validation required: YES

Notes: Monthly/region-specific

[Cost C013 - Transport to care](#)

Category: access

Cost level: C1-C3

Min EUR: 5

Max EUR: 200

Public cost possible: YES

Private cost possible: YES

Pathway: PATH_TRANSPORT_ACCESS

Fallback cost note: Telemedicine/local alternative

Local validation required: YES

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

[Cost C014 - Document/scanning/printing/postage](#)

Category: documental

Cost level: C0-C1

Min EUR: 0

Max EUR: 50

Public cost possible: NO

Private cost possible: YES

Pathway: PATH_PUBLIC_DUTY

Fallback cost note: Digital where possible

Local validation required: YES

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

[Cost C015 - Psychological session/support](#)

Category: mental

Cost level: C1-C3

Min EUR: 0

Max EUR: 100

Public cost possible: YES

Private cost possible: YES

Pathway: PATH_MENTAL_HEALTH

Fallback cost note: CSM/public/association first

Local validation required: YES

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

[Cost C016 - CSM/urgent mental health route](#)

Category: mental

Cost level: C0-C1

Min EUR: 0

Max EUR: 50

Public cost possible: YES

Private cost possible: NO/rare

Pathway: PATH_MENTAL_HEALTH

Fallback cost note: Emergency if safety trigger

Local validation required: YES

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

[Cost C017 - Social worker assessment](#)

Category: social

Cost level: C0-C1

Min EUR: 0

Max EUR: 50

Public cost possible: YES

Private cost possible: NO/rare

Pathway: PATH_ECONOMIC_RESCUE

Fallback cost note: Public/municipal

Local validation required: YES

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

[Cost C018 - Legal access/public duty dossier](#)

Category: legal

Cost level: C0-C2

Min EUR: 0

Max EUR: 300

Public cost possible: YES/NGO

Private cost possible: YES

Pathway: PATH_LEGAL_COLLAPSE_REVIEW

Fallback cost note: Template generated by H1

Local validation required: YES

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

[Cost C019 - Grant/funding application preparation](#)

Category: funding

Cost level: C0-C3

Min EUR: 0

Max EUR: 1000

Public cost possible: YES

Private cost possible: YES

Pathway: PATH_FUNDING_REQUEST

Fallback cost note: Template minimizes cost

Local validation required: YES

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

[Cost C020 - Wet-lab/preclinical validation](#)

Category: research lab

Cost level: C5

Min EUR: 5000

Max EUR: 1000000

Public cost possible: YES grant

Private cost possible: YES

Pathway: PATH_EDEN_LAB_HANDOFF

Fallback cost note: Only authorized lab/institution

Local validation required: YES

Notes: No operational wet-lab protocol included

5. Canonical register 3 - Role Execution Map

The role map defines who can do what, what cannot be done, response times, required documents and escalation when a role fails or blocks the pathway.

Role R001 - AI Guardian

Permissions: calculate, check formulas, flag safety/capture, draft documents

Forbidden actions: diagnosis, prescription, wet-lab execution

Material actions: Generate pathways, reports, dossiers

Target response time: Immediate

Fallback: Human authorized review where legally needed

Escalation: Founder review for kernel/brand

Required documents: Version log

Audit required: YES

Role R002 - Founder Steward

Permissions: approve kernel/brand/mission changes

Forbidden actions: clinical decisions, data falsification

Material actions: Approve strategic releases and anti-capture compliance

Target response time: Strategic

Fallback: AI Guardian flag

Escalation: External legal/technical audit

Required documents: Approval log

Audit required: YES

Role R003 - Case Coordinator

Permissions: intake, collect data, route pathways

Forbidden actions: diagnose/prescribe, alter scores without log

Material actions: Complete forms, trigger referrals, track outcomes

Target response time: 48h for critical case

Fallback: Supervisor/AI Guardian

Escalation: Public responsibility if blocked

Required documents: Consent, intake, baseline

Audit required: YES

Role R004 - Clinician

Permissions: clinical assessment, diagnosis, prescription

Forbidden actions: economic gatekeeping, unlogged refusal

Material actions: Clinical referral, treatment plan, urgent triage

Target response time: 48h/7d according to severity

Fallback: Second clinician/center

Escalation: Health authority

Required documents: Clinical records

Audit required: YES

Role R005 - Mental Health Professional

Permissions: mental assessment/support

Forbidden actions: ignore suicide flag, unlogged decision

Material actions: PHQ/GAD review, CSM referral, therapy plan

Target response time: 48h critical / 7d standard

Fallback: Emergency route

Escalation: Health authority

Required documents: Mental screening

Audit required: YES

Role R006 - Social Worker

Permissions: economic/social assessment, public support

Forbidden actions: clinical diagnosis, blocking support for irrelevant reasons

Material actions: Housing/food/benefit activation

Target response time: 7d critical economic

Fallback: Municipality/foundation bridge

Escalation: Regional/State escalation

Required documents: EFS, documents

Audit required: YES

[Role R007 - Authorized Lab](#)

Permissions: authorized tests/validation

Forbidden actions: unapproved human experimentation

Material actions: Lab testing, validation report

Target response time: per protocol

Fallback: Alternative lab

Escalation: Regulatory body

Required documents: Lab requisition/protocol

Audit required: YES

[Role R008 - University/Research Institute](#)

Permissions: validation, Delphi, research protocol

Forbidden actions: claim clinical efficacy without data

Material actions: External validation, pilot design

Target response time: 30-90d

Fallback: Second institute

Escalation: Public grant route

Required documents: Research protocol

Audit required: YES

[Role R009 - Public Actor](#)

Permissions: fund/enable public services

Forbidden actions: non-response without record

Material actions: Funding, access, duty response

Target response time: 30d standard / urgent for critical

Fallback: Escalation dossier

Escalation: Higher authority/media/legal

Required documents: Public dossier

Audit required: YES

Role R010 - Foundation/Funder

Permissions: fund low-cost access/pilot

Forbidden actions: capture kernel, predatory control

Material actions: Bridge funds, pilot grants

Target response time: 30d

Fallback: Public funding/other foundation

Escalation: Transparency log

Required documents: Funding request

Audit required: YES

Role R011 - Legal/Policy Analyst

Permissions: LCIS, procedural harm, public duty memo

Forbidden actions: clinical substitution

Material actions: Legal collapse review, templates

Target response time: 14-30d

Fallback: Public actor route

Escalation: Court/ombudsman/media as applicable

Required documents: Evidence dossier

Audit required: YES

6. Canonical register 4 - Physical Inventory

The physical inventory lists digital, clinical, documentary and specialist materials required to make the system activable. Each item includes tier, minimum quantity, cost, low-cost substitute and safety requirement.

Item I001 - Laptop or desktop

Tier: Tier0

Minimum quantity: 1

Estimated cost level: C2-C3

Low-cost substitute: Shared workstation

Public provider: Public office/university

Private provider: Retail/refurbished

Safety requirement: Password, encryption

Used in pathways: All

Notes: Refurbished acceptable

Item I002 - Smartphone

Tier: Tier0

Minimum quantity: 1

Estimated cost level: C1-C3

Low-cost substitute: Shared phone/clinic phone

Public provider: Public service

Private provider: Retail

Safety requirement: Secure messaging policy

Used in pathways: All

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

Item I003 - Scanner/app scanner

Tier: Tier0

Minimum quantity: 1

Estimated cost level: C0-C1

Low-cost substitute: Phone camera

Public provider: Public library/office

Private provider: Retail/app

Safety requirement: Avoid leaking sensitive data

Used in pathways: Dossier generation

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

Item I004 - Printer/postage access

Tier: Tier0

Minimum quantity: 1

Estimated cost level: C0-C1

Low-cost substitute: Digital submission

Public provider: Public office/library

Private provider: Retail

Safety requirement: Document privacy

Used in pathways: Public duty/legal

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

[Item I005 - Secure cloud/storage](#)

Tier: Tier0

Minimum quantity: 1

Estimated cost level: C0-C2

Low-cost substitute: Encrypted local storage

Public provider: Institutional IT

Private provider: Commercial cloud

Safety requirement: Encryption, access control

Used in pathways: All

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

[Item I006 - Blood pressure monitor](#)

Tier: Tier1

Minimum quantity: 1

Estimated cost level: C1-C2

Low-cost substitute: Pharmacy/public measurement

Public provider: Clinic/pharmacy

Private provider: Retail

Safety requirement: Calibrated if possible

Used in pathways: Longevity/clinical

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

Item I007 - Scale

Tier: Tier1

Minimum quantity: 1

Estimated cost level: C1

Low-cost substitute: Pharmacy measurement

Public provider: Clinic/pharmacy

Private provider: Retail

Safety requirement: Privacy

Used in pathways: Longevity

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

Item I008 - Pulse oximeter optional

Tier: Tier1

Minimum quantity: 1

Estimated cost level: C1

Low-cost substitute: Clinical measurement

Public provider: Clinic

Private provider: Retail

Safety requirement: Not diagnostic alone

Used in pathways: Clinical triage optional

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

Item I009 - Questionnaire pack PHQ/GAD/NRS/UCLA/MLQ/LOT-R

Tier: Tier1

Minimum quantity: 1

Estimated cost level: C0

Low-cost substitute: Printed templates

Public provider: Project repository

Private provider: N/A

Safety requirement: Use scoring rules

Used in pathways: Mental/Happiness

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

[Item I010 - Basic clinical laboratory access](#)

Tier: Tier1

Minimum quantity: 1

Estimated cost level: C1-C2

Low-cost substitute: Public labs

Public provider: SSN/public system

Private provider: Private lab

Safety requirement: Authorized laboratory

Used in pathways: Longevity/clinical

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

[Item I011 - Genetic testing access](#)

Tier: Tier2

Minimum quantity: 0

Estimated cost level: C3-C5

Low-cost substitute: Only if clinically indicated

Public provider: Genetics center

Private provider: Authorized lab

Safety requirement: Consent explicit

Used in pathways: Advanced therapy

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

[Item I012 - Biological age clocks access](#)

Tier: Tier2

Minimum quantity: 0

Estimated cost level: C3-C4

Low-cost substitute: Not required for L0

Public provider: Research center

Private provider: Provider

Safety requirement: Consent, DQL flag

Used in pathways: Longevity advanced

Notes: Optional

[Item I013 - Trial registry access](#)

Tier: Tier2

Minimum quantity: 1

Estimated cost level: C0

Low-cost substitute: Public web search

Public provider: ClinicalTrials/EU CTR

Private provider: N/A

Safety requirement: Verify source

Used in pathways: Trial route

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

[Item I014 - EDEN/AI hypothesis access](#)

Tier: Tier3

Minimum quantity: 0

Estimated cost level: C0-C5

Low-cost substitute: Literature/partner review

Public provider: University/lab

Private provider: Commercial AI/research

Safety requirement: Hypothesis only, no clinical use

Used in pathways: EDEN/Lab

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

Item I015 - Authorized lab partner

Tier: Tier3

Minimum quantity: 0

Estimated cost level: C5

Low-cost substitute: University collaboration

Public provider: Public research institute

Private provider: Contract research org

Safety requirement: Ethics/regulatory approval

Used in pathways: Lab route

Notes: local validation required. The cost range is provisional and not an official tariff; use public or low-cost alternatives whenever available.

7. Canonical register 5 - Pathway Library

The pathway library contains executable time-sequenced pathways. Each pathway indicates steps within 48 hours, 7 days, 30 days, 90 days, documents, roles, costs, fallback, final output and outcome metric.

Pathway PATH_PAIN_48H - Pain Urgent Pathway

Problem: PAIN_SEVERE

Trigger: NRS $\geq$ 7

Step 1 - 0/48h: Confirm NRS, red flags, current meds; safety check

Step 2 - 7 days: Contact clinician/pain service; schedule evaluation

Step 3 - 30 days: Track NRS, sleep, function; activate therapy/fisio route

Step 4 - 90 days: Outcome review; escalate if NRS not improved

Documents: Pain log, medication list, prior exams

Roles: Case Coordinator; Clinician; Mental Pro if needed

Costs: C003,C004,C015

Fallback 1: Alternative pain center

Fallback 2: Public duty if severe pain untreated

Final output: CLINICAL_ROUTE

Outcome metric: NRS delta, function, sleep

Pathway PATH_MEDICATION_ACCESS - Medication Access Pathway

Problem: MEDICATION_MISSING

Trigger: Essential medication unavailable

Step 1 - 0/48h: Identify medication, urgency, reason unavailable

Step 2 - 7 days: Contact prescriber/pharmacy; check generic/exemption/fund

Step 3 - 30 days: Stabilize supply, document cost barrier

Step 4 - 90 days: Outcome: continuous access or escalation

Documents: Prescription, diagnosis, pharmacy cost

Roles: Case Coordinator; Clinician; Social Worker

Costs: C001,C014

Fallback 1: Economic Rescue

Fallback 2: Public Responsibility

Final output: SOLVE_NOW

Outcome metric: Medication available yes/no

Pathway PATH_ECONOMIC_RESCUE - Economic Rescue Pathway

Problem: ECONOMIC_COLLAPSE

Trigger: EFS<=14

Step 1 - 0/48h: Run H1-EFS; identify critical items housing/food/meds

Step 2 - 7 days: Activate social services/benefits/food/housing/funds

Step 3 - 30 days: Check EFS change and unresolved barriers

Step 4 - 90 days: Stabilize plan or escalate public duty

Documents: EFS, income/cost docs if available

Roles: Case Coordinator; Social Worker; Public Actor

Costs: C011,C012,C017

Fallback 1: Foundation bridge

Fallback 2: Public Responsibility

Final output: ECONOMIC_RESCUE

Outcome metric: EFS delta; critical flags resolved

Pathway PATH_HOUSING_RESCUE - Housing Rescue Pathway

Problem: HOUSING_CRITICAL

Trigger: EFS_2<=1

Step 1 - 0/48h: Confirm housing risk and safety

Step 2 - 7 days: Contact municipal/emergency housing/social support

Step 3 - 30 days: Bridge fund or safe temporary plan

Step 4 - 90 days: Longer-term housing stabilization

Documents: Housing notice, ID, EFS

Roles: Social Worker; Public Actor; Foundation

Costs: C012,C014

Fallback 1: Foundation/charity route

Fallback 2: State escalation

Final output: ECONOMIC_RESCUE

Outcome metric: Housing stable yes/no

Pathway PATH_FOOD_SECURITY - Food Security Pathway

Problem: FOOD_CRITICAL

Trigger: EFS_3<=1

Step 1 - 0/48h: Confirm food insecurity

Step 2 - 7 days: Connect food bank/municipal vouchers

Step 3 - 30 days: Check food stability and nutrition adequacy

Step 4 - 90 days: Integrate income/benefit plan

Documents: EFS, household info

Roles: Social Worker; Foundation

Costs: C011

Fallback 1: Community partner

Fallback 2: Public Responsibility

Final output: ECONOMIC_RESCUE

Outcome metric: Food security score

Pathway PATH_CLINICAL_REFERRAL - Clinical Referral Pathway

Problem: CARE_ACCESS_BLOCKED

Trigger: A_raw>=4

Step 1 - 0/48h: Identify missing visit/exam and urgency

Step 2 - 7 days: Request appointment/referral; check alternate centers

Step 3 - 30 days: Confirm appointment completed/pending

Step 4 - 90 days: Escalate if blocked

Documents: Referral, symptoms, urgency note

Roles: Case Coordinator; Clinician

Costs: C002,C003,C014

Fallback 1: Alternative center

Fallback 2: Public Responsibility

Final output: CLINICAL_ROUTE

Outcome metric: Visit/exam obtained

Pathway PATH_MENTAL_HEALTH - Mental Health Pathway

Problem: DEPRESSION_HIGH

Trigger: PHQ-9>=15 or item 9 positive

Step 1 - 0/48h: Safety screen; urgent route if self-harm risk

Step 2 - 7 days: Clinician/CSM/psychological referral; address root causes

Step 3 - 30 days: Repeat PHQ/GAD; check social/economic drivers

Step 4 - 90 days: Outcome review and long-term plan

Documents: PHQ-9,GAD-7,UCLA,EFS,NRS

Roles: Case Coordinator; Mental Pro; Clinician

Costs: C015,C016

Fallback 1: Emergency route

Fallback 2: Public duty/social route

Final output: CLINICAL_ROUTE

Outcome metric: PHQ/GAD delta, safety

Pathway PATH_ANTI_ISOLATION - Anti-Isolation Pathway

Problem: ISOLATION_HIGH

Trigger: UCLA high

Step 1 - 0/48h: Identify social isolation source

Step 2 - 7 days: Connect community/volunteer/family/social services

Step 3 - 30 days: Track contact frequency and UCLA/support score

Step 4 - 90 days: Sustain social link / escalate if unresolved

Documents: UCLA, support map

Roles: Case Coordinator; Social Worker

Costs: C014,C017

Fallback 1: Community partner

Fallback 2: Public Responsibility

Final output: SOLVE_NOW

Outcome metric: UCLA/support delta

Pathway PATH_LONGEVITY_BASELINE - Longevity Prevention Baseline

Problem: AGING_RISK_HIGH

Trigger: function/biomarker risk

Step 1 - 0/48h: Collect baseline function/labs if indicated

Step 2 - 7 days: Implement L0 plan: movement, sleep, nutrition, medical follow-up

Step 3 - 30 days: Repeat function/biomarkers where feasible

Step 4 - 90 days: Outcome review 90/180/365

Documents: Clinical/lab results, function tests

Roles: Clinician; Authorized Lab; Case Coord

Costs: C005,C006,C007

Fallback 1: Clinical preventive route

Fallback 2: Research route

Final output: CLINICAL_ROUTE

Outcome metric: Function/biomarker delta

Pathway PATH_ADVANCED_THERAPY_ACCESS - Advanced Therapy Access

Problem: ADVANCED_THERAPY_INACCESSIBLE

Trigger: Approved therapy exists but inaccessible

Step 1 - 0/48h: Confirm indication and therapy status

Step 2 - 7 days: Collect eligibility docs; identify center and coverage

Step 3 - 30 days: Submit referral/funding request

Step 4 - 90 days: Track access/funding/center response

Documents: Diagnosis, genetic/clinical docs, coverage docs

Roles: Clinician; Specialist; Public Actor

Costs: C009,C010,C019

Fallback 1: Funding route

Fallback 2: Public responsibility

Final output: FUNDING_ROUTE

Outcome metric: Eligibility/referral status

Pathway PATH_GENE_CELL_ACCESS - Gene/Cell Therapy Access

Problem: GENETIC_DISEASE_TARGET

Trigger: diagnosis may match therapy/trial

Step 1 - 0/48h: Confirm diagnosis and indication

Step 2 - 7 days: Identify approved therapy/trial and required tests

Step 3 - 30 days: Referral to specialist/center; funding path

Step 4 - 90 days: Outcome: eligibility/access/trial status

Documents: Genetic report, clinical record

Roles: Clinician; Genetics center; Lab

Costs: C009,C010

Fallback 1: Trial route

Fallback 2: Research/public funding

Final output: TRIAL_ROUTE

Outcome metric: Eligibility verified

[Pathway PATH_EDEN_LAB_HANDOFF - EDEN/Lab Invention Handoff](#)

Problem: UNRESOLVED_BIOLOGY

Trigger: no adequate solution

Step 1 - 0/48h: Define unresolved mechanism and target

Step 2 - 7 days: Prepare hypothesis/lab handoff and partner map

Step 3 - 30 days: Submit grant/proposal to authorized lab

Step 4 - 90 days: Track validation result

Documents: Hypothesis brief, evidence, safety boundary

Roles: Research Institute; Authorized Lab

Costs: C019,C020

Fallback 1: Grant concept

Fallback 2: Invention backlog

Final output: LAB_ROUTE

Outcome metric: Hypothesis accepted/validated

[Pathway PATH_LEGAL_COLLAPSE_REVIEW - Legal Collapse Impact Review](#)

Problem: LEGAL_ACCESS_HARM

Trigger: law/procedure blocks access

Step 1 - 0/48h: Document harm and affected capability

Step 2 - 7 days: Prepare LCIS and public duty memo

Step 3 - 30 days: Submit to relevant actor; log response

Step 4 - 90 days: Escalate non-response

Documents: Evidence, timeline, harm statement

Roles: Legal Analyst; Public Actor

Costs: C014,C018

Fallback 1: Escalation dossier

Fallback 2: Higher authority

Final output: LEGAL_ROUTE

Outcome metric: Barrier removed/response

Pathway PATH_FUNDING_REQUEST - Funding Request Pathway

Problem: FUNDING_GAP

Trigger: solution exists but no payer

Step 1 - 0/48h: Calculate cost and cost of inaction

Step 2 - 7 days: Create funding dossier; identify payer/foundation/grant

Step 3 - 30 days: Submit and track response

Step 4 - 90 days: Escalate if unresolved

Documents: Cost table, problem, outcome target

Roles: Case Coord; Foundation; Public Actor

Costs: C019

Fallback 1: Public duty

Fallback 2: Foundation/UE grant

Final output: FUNDING_ROUTE

Outcome metric: Funding secured yes/no

Pathway PATH_TRANSPORT_ACCESS - Transport Access Pathway

Problem: TRANSPORT_BARRIER

Trigger: cannot reach service

Step 1 - 0/48h: Confirm transport barrier

Step 2 - 7 days: Arrange transport/telemedicine/local alternative

Step 3 - 30 days: Check attendance/completion

Step 4 - 90 days: Integrate ongoing support

Documents: Appointment, cost, transport needs

Roles: Social Worker; Case Coord

Costs: C013

Fallback 1: Local partner

Fallback 2: Public duty

Final output: SOLVE_NOW

Outcome metric: Appointment reached yes/no

Pathway PATH_DIGITAL_SUPPORT - Digital Access Support

Problem: DIGITAL_BARRIER

Trigger: cannot use portals/forms

Step 1 - 0/48h: Identify portal/document barrier

Step 2 - 7 days: Provide assisted digital submission or paper route

Step 3 - 30 days: Confirm application submitted

Step 4 - 90 days: Monitor response

Documents: Documents, consent

Roles: Case Coord; Social Worker

Costs: C014

Fallback 1: Paper route

Fallback 2: Public service escalation

Final output: SOLVE_NOW

Outcome metric: Submission completed

Pathway PATH_DEBT_HEALTH_STABILIZATION - Debt/Health Cost Stabilization

Problem: DEBT_HEALTH_BLOCK

Trigger: EFS_6<=1

Step 1 - 0/48h: Quantify debt/health cost barrier

Step 2 - 7 days: Contact social/legal/debt support; seek moratorium/fund

Step 3 - 30 days: Track essential access restored

Step 4 - 90 days: Longer-term stabilization

Documents: Bills, debt summary, EFS

Roles: Social Worker; Legal Analyst

Costs: C014,C018

Fallback 1: Foundation bridge

Fallback 2: Public duty

Final output: ECONOMIC_RESCUE

Outcome metric: EFS_6 delta

Pathway PATH_MEANING_FUTURE - Meaning/Future Pathway

Problem: LOW_MEANING_FUTURE

Trigger: MLQ/LOT-R low

Step 1 - 0/48h: Measure MLQ/LOT-R and root causes

Step 2 - 7 days: Build 30-day meaningful activity + social/economic actions

Step 3 - 30 days: Repeat scales and adjust plan

Step 4 - 90 days: Sustain project of life

Documents: MLQ, LOT-R, PHQ/GAD, EFS

Roles: Case Coord; Mental Pro; Social Worker

Costs: C015,C017

Fallback 1: Social connection

Fallback 2: Economic/health route

Final output: SOLVE_NOW

Outcome metric: MLQ/LOT-R delta

8. Pre-filled Material Solution Units

Material Solution Units are the minimum material cell of the system: problem, trigger, cause, solution, physical component, digital component, authorized component, cost, funding, access, time, metric and fallback.

MSU MSU001 - P001

Domain: PAIN

Trigger: NRS $\geq$ 7

Cause: Untreated pain/access barrier

Solution type: clinical service

Material component: Pain specialist/clinician + therapy plan

Digital component: Pain tracker + dossier

Authorized component: Clinician/pain center

Estimated cost: C1-C3

Funding: SSN/fund/private bridge

Access: PATH_PAIN_48H

Timeline: 48h/7d

Outcome metric: NRS delta; function

Fallback 1: Alternative center

Fallback 2: Public duty

Status: planned

[MSU MSU002 - P002](#)

Domain: ACCESS/ECON

Trigger: essential meds missing

Cause: Cost/prescription/access

Solution type: medication access

Material component: Medication supply/exemption/fund

Digital component: Medication access log

Authorized component: Clinician/pharmacy/social worker

Estimated cost: C1-C3

Funding: SSN/exemption/fund

Access: PATH_MEDICATION_ACCESS

Timeline: 48h

Outcome metric: medication available

Fallback 1: Economic rescue

Fallback 2: Public duty

Status: planned

[MSU MSU003 - P009](#)

Domain: ECON

Trigger: EFS<=14

Cause: Material insecurity

Solution type: economic rescue

Material component: Housing/food/medication/care support

Digital component: EFS + rescue plan

Authorized component: Social worker/public actor

Estimated cost: C1-C5

Funding: Public/foundation/municipality

Access: PATH_ECONOMIC_RESCUE

Timeline: 7d/30d

Outcome metric: EFS delta

Fallback 1: Foundation bridge

Fallback 2: Public duty

Status: planned

[MSU MSU004 - P006](#)

Domain: MENTAL

Trigger: PHQ-9 $\geq$ 15

Cause: Depression/root causes

Solution type: mental health route

Material component: CSM/clinician/psychological support

Digital component: Mental pathway tracker

Authorized component: Mental health professional

Estimated cost: C0-C3

Funding: SSN/fund

Access: PATH_MENTAL_HEALTH

Timeline: 48h/7d

Outcome metric: PHQ/GAD delta

Fallback 1: Emergency route

Fallback 2: Social/economic root cause

Status: planned

[MSU MSU005 - P010](#)

Domain: LONGEVITY

Trigger: aging risk

Cause: metabolic/inflammatory/function

Solution type: prevention package

Material component: basic labs, movement, sleep, nutrition

Digital component: Longevity baseline workbook

Authorized component: Clinician/lab

Estimated cost: C1-C3

Funding: SSN/private low-cost/grant

Access: PATH_LONGEVITY_BASELINE

Timeline: 30d/90d

Outcome metric: function/biomarker delta

Fallback 1: Clinical preventive route

Fallback 2: Research route

Status: planned

[MSU MSU006 - P011](#)

Domain: ADVANCED_ACCESS

Trigger: therapy exists but inaccessible

Cause: cost/eligibility/center

Solution type: advanced therapy access

Material component: referral, documents, funding route

Digital component: Advanced therapy access map

Authorized component: Specialist/center/public actor

Estimated cost: C4-C5

Funding: Public/insurance/foundation/grant

Access: PATH_ADVANCED_THERAPY_ACCESS

Timeline: 30d/90d

Outcome metric: eligibility/referral status

Fallback 1: Funding route

Fallback 2: Public duty

Status: planned

[MSU MSU007 - P014](#)

Domain: RESEARCH

Trigger: unresolved biology

Cause: scientific gap

Solution type: lab handoff

Material component: hypothesis brief + partner lab

Digital component: EDEN hypothesis log

Authorized component: Authorized lab/university

Estimated cost: C5

Funding: Grant/foundation/state

Access: PATH_EDEN_LAB_HANDOFF

Timeline: 90d/12m

Outcome metric: validation accepted/result

Fallback 1: Grant concept

Fallback 2: Invention backlog

Status: planned

[MSU MSU008 - P015](#)

Domain: LEGAL

Trigger: procedure blocks access

Cause: legal/procedural harm

Solution type: legal-state action

Material component: LCIS + public responsibility dossier

Digital component: Legal collapse template

Authorized component: Legal analyst/public actor

Estimated cost: C0-C2

Funding: public/legal aid/project

Access: PATH_LEGAL_COLLAPSE_REVIEW

Timeline: 30d

Outcome metric: barrier response/removal

Fallback 1: Escalation dossier

Fallback 2: Higher authority

Status: planned

9. Three complete demonstrative cases

The three demonstrative cases show the transition from intake to baseline, pathway, material action and 30-day outcome. They are not external clinical validation; they are synthetic operational cases to verify that the factory produces a complete pathway.

Case CASE_PAIN

Phase: Intake

Problem: PAIN_SEVERE

Key inputs: NRS=8; visit missing; EFS=22

Initial/current K: 52

Initial/current HFI: 48

Initial/current EFS: 22

Triggered pathway: PATH_PAIN_48H

Material action: Pain referral + pain log + access check

Authorized actor: Clinician/pain center

Cost route: SSN/bridge fund

t30 outcome: NRS=5; visit scheduled; sleep improved

t30 RRS: 0.66

Next step: Continue 90d follow-up

Phase: Baseline

Problem: PAIN_SEVERE

Key inputs: B=2 M=3 E=2 A=4 S=2

Initial/current K: 52

Initial/current HFI: 48

Initial/current EFS: 22

Triggered pathway: PATH_PAIN_48H

Material action: Pain urgent pathway active

Authorized actor: Case coord + clinician

Cost route: C003/C004

t30 outcome:

t30 RRS:

Next step:

Phase: Outcome t30

Problem: PAIN_SEVERE

Key inputs: NRS 8→5; A 4→2

Initial/current K: 38

Initial/current HFI: 58

Initial/current EFS: 24

Triggered pathway: PATH_PAIN_48H

Material action: Partial material resolution

Authorized actor: Clinician

Cost route: SSN

t30 outcome: Pain reduced ≥ 2 points

t30 RRS: 0.66

Next step: Monitor/optimize

Case CASE_ECON

Phase: Intake

Problem: ECONOMIC_COLLAPSE

Key inputs: EFS=9; housing=1; food=2; meds=1

Initial/current K: 63

Initial/current HFI: 35

Initial/current EFS: 9

Triggered pathway: PATH_ECONOMIC_RESCUE

Material action: Food/meds/housing emergency requests

Authorized actor: Social worker/public actor

Cost route: Municipal/foundation

t30 outcome: EFS=18; meds available; food stable

t30 RRS: 0.72

Next step: Public duty for housing

Phase: Baseline

Problem: ECONOMIC_COLLAPSE

Key inputs: B=2 M=3 E=5 A=3 S=3

Initial/current K: 63

Initial/current HFI: 35

Initial/current EFS: 9

Triggered pathway: PATH_ECONOMIC_RESCUE

Material action: Economic rescue active

Authorized actor: Social worker

Cost route: C011/C012/C017

t30 outcome:

t30 RRS:

Next step:

Phase: Outcome t30

Problem: ECONOMIC_COLLAPSE

Key inputs: EFS 9→18; medication resolved

Initial/current K: 48

Initial/current HFI: 49

Initial/current EFS: 18

Triggered pathway: PATH_ECONOMIC_RESCUE

Material action: Partial material stabilization

Authorized actor: Social worker/public actor

Cost route: Public/foundation

t30 outcome: EFS improved +9; meds yes; housing pending

t30 RRS: 0.72

Next step: Escalate housing

Case CASE_ADV

Phase: Intake

Problem: ADVANCED_THERAPY_INACCESSIBLE

Key inputs: Therapy exists; cost/center barrier

Initial/current K: 70

Initial/current HFI: 42

Initial/current EFS: 16

Triggered pathway: PATH_ADVANCED_THERAPY_ACCESS

Material action: Eligibility docs + funding dossier

Authorized actor: Specialist/public actor

Cost route: Public/foundation/grant

t30 outcome: Referral submitted; funding pending

t30 RRS: 0.41

Next step: Public responsibility

Phase: Baseline

Problem: ADVANCED_THERAPY_INACCESSIBLE

Key inputs: B=4 M=2 E=4 A=5 S=2

Initial/current K: 70

Initial/current HFI: 42

Initial/current EFS: 16

Triggered pathway: PATH_ADVANCED_THERAPY_ACCESS

Material action: Advanced therapy access mapping

Authorized actor: Specialist/center

Cost route: C009/C010/C019

t30 outcome:

t30 RRS:

Next step:

Phase: Outcome t30

Problem: ADVANCED_THERAPY_INACCESSIBLE

Key inputs: Center identified; documents 70% complete

Initial/current K: 66

Initial/current HFI: 45

Initial/current EFS: 16

Triggered pathway: PATH_ADVANCED_THERAPY_ACCESS

Material action: Access pathway active not resolved

Authorized actor: Specialist/public actor

Cost route: Funding route

t30 outcome: Eligibility review scheduled; cost unresolved

t30 RRS: 0.41

Next step: Funding escalation

10. Calculated Real Resolution Score

The Real Resolution Score measures how close a problem is to material resolution. The formula uses seven components: problem detected, causal clarity, solution available, physical deliverability, funding access, authorized execution and outcome improvement. The final score ranges from 0 to 1.

RRS CASE_PAIN

ProblemDetected: 1

CausalClarity: 0.8

SolutionAvailable: 0.8

PhysicalDeliverability: 0.7

FundingAccess: 0.6

AuthorizedExecution: 0.7

OutcomeImprovement: 0

Calculated RRS: 0.705

Status: pending

[RRS CASE_ECON](#)

ProblemDetected: 1

CausalClarity: 0.9

SolutionAvailable: 0.8

PhysicalDeliverability: 0.8

FundingAccess: 0.6

AuthorizedExecution: 0.8

OutcomeImprovement: 0.2

Calculated RRS: 0.765

Status: active

[RRS CASE_ADV_ACCESS](#)

ProblemDetected: 1

CausalClarity: 0.7

SolutionAvailable: 0.7

PhysicalDeliverability: 0.4

FundingAccess: 0.2

AuthorizedExecution: 0.5

OutcomeImprovement: 0

Calculated RRS: 0.550

Status: funding required

11. Calculated Cost of Inaction

The Cost of Inaction is not a final accounting estimate. It is an operational index documenting the foreseeable harm of non-intervention. It converts public or institutional delay into measurable responsibility.

[COI_PAIN](#)

Case: Untreated severe pain

Calculated index: 0.76

Class: high

health deterioration: 5

function loss: 4

mental burden: 4

economic spillover: 3

future service use: 3

COI_ECON

Case: Economic collapse with housing/food/medication risk

Calculated index: 0.92

Class: critical

health access: 5

housing risk: 5

food/medication: 5

mental health: 4

social collapse: 4

COI_ADV

Case: Existing advanced therapy but inaccessible

Calculated index: 0.96

Class: critical

health loss: 5

irreversibility: 5

equity failure: 5

cost barrier: 5

public mandate: 4

12. Populated dossiers and material outputs

This section contains pre-populated examples of public dossiers, funding request, laboratory handoff and legal collapse impact. They are operational models to adapt to the actual case and competent jurisdiction.

PublicResponsibilityDossierSeverePain

Subject: Untreated severe pain

Situation: The person presents severe pain with NRS equal to or above 7, delayed specialist access and risk of functional and mental deterioration.

Concrete request: Access to clinical evaluation and pain care within 48 hours or 7 days according to severity.

Attachable documents: NRS, pain log, medication list, missed visits, available reports.

Required outcome: NRS reduction, sleep and function improvement, clinical pathway activation.

Suggested response deadline: 14-30 days according to urgency

If no response: Non-Response Accountability Log + escalation

PublicResponsibilityDossierEconomicCollapse

Subject: Economic collapse with housing/food/medication risk

Situation: The person has EFS equal to or below 14, with material risk compromising health, access to medication and housing stability.

Concrete request: Immediate activation of economic support, food vouchers, medication fund, housing support and social assistance.

Attachable documents: EFS, cost documents, medication, housing, income where available.

Required outcome: EFS increased, medication available, food stable, housing risk reduced.

Suggested response deadline: 14-30 days according to urgency

If no response: Non-Response Accountability Log + escalation

PublicResponsibilityDossierAdvancedTherapy

Subject: Existing advanced therapy but inaccessible

Situation: A specialist or advanced solution may exist, but access, cost or eligibility are unresolved.

Concrete request: Eligibility check, authorized center, clinical documents, funding route and response from competent actor.

Attachable documents: Diagnosis, reports, indicated tests, clinical history, estimated costs.

Required outcome: Eligibility verified, referral activated, funding request or public duty documented.

Suggested response deadline: 14-30 days according to urgency

If no response: Non-Response Accountability Log + escalation

LabHandoffInaccessibleAdvancedTherapy

Purpose: Transfer to a laboratory, university or authorized center a case where the solution requires specialist validation or an advanced pathway.

Does not contain: Wet-lab procedures, genetic editing, cellular manipulation or experimental instructions.

Contains: Problem, clinical/scientific target, available evidence, required data, eligibility criteria, validation gate, expected outcome.

Authorized actor: Laboratory, university, clinical center or other authorized structure.

Output: Accepted for evaluation, additional data requested, rejected with reason, or routed to trial/competent center.

FundingRequestEconomicCollapse

Subject: Bridge funding request for economic collapse with health impact.

Basis: EFS equal to or below 14; risk affecting medication, food, housing or care.

Request: Temporary coverage for medication, food, health transport, essential visits, housing stabilization.

Reason: Non-intervention increases K, reduces HFI and produces higher future public cost.

Outcome: EFS increased, medication guaranteed, care access restored, housing/food risk reduced.

LegalCollapseImpactDeniedAccess

Subject: Impact of a procedure or practice producing denied access.

Harm: Delay, economic barrier, non-response or opaque procedure worsening health or material security.

Criterion: If a solution exists but does not reach the person because of public, economic or procedural barrier, the case becomes a Public Responsibility Case.

Request: Procedural correction, reasoned response, alternative access, funding or referral.

Legal-operational outcome: Barrier removed, responsible actor identified, escalation documented.

13. A-Z technical manual

1. Open the central workbook and create a new problem_id.
2. Check whether the problem exists in the Problem Catalog; if not, create a provisional entry with severity, urgency and target factory.
3. Complete the Material Solution Unit with trigger, cause, material component, digital component, authorized actor, cost, funding, access, timeline and outcome.
4. Consult the Pathway Library and activate the primary pathway.
5. If the pathway requires an authorized role, use the Role Execution Map and register the maximum response time.
6. Consult the Cost Table and select public, private, foundation or funding route.
7. If cost blocks the solution, activate Funding Engine and Public Responsibility Dossier.

8. If the solution requires laboratory or trial, generate Lab Handoff without unauthorized procedures.
9. Calculate Real Resolution Score before and after the intervention.
10. Record 30-day and 90-day outcomes.
11. If the problem is not resolved, activate fallback 1; if it fails, fallback 2; if it still fails, record Public Failure or Invention Route.
12. No case is closed without outcome, cost, pathway, responsible actor and final state.

Boundaries

The system does not authorize experimentation, therapies, prescriptions or biological manipulations. Where clinical competences, laboratories or institutions are required, the system produces handoff, dossier, request, tracking and outcome, but does not replace authorized actors.

14. Operational conclusion

With the integration of canonical registers and pre-filled cases, HUMANITY-ONE Material Solutions is no longer only a technical report. It becomes a documentary and operational factory: it identifies the problem, selects the factory, produces an MSU, activates a pathway, assigns roles, estimates costs, produces fallbacks, calculates RRS and generates responsibility dossiers when the public or institutional system does not respond.

For use, implementation, adaptation, validation or written authorization: pierernesto.di.paolo.69@proton.me

15. Extended technical operating manual

This extended operating manual completes the technical report and must be read as the execution guide for the HUMANITY-ONE Material Solutions package. Its purpose is not to describe an abstract principle, but to explain how an institution, authorized group or partner laboratory should open the package, identify the problem, select the pathway, activate the material solution, measure the outcome and record failures or public responsibilities.

15.1 Package preparation

- Create a local folder named HUMANITY_ONE_MATERIAL_SOLUTIONS and keep one unmodified copy as the original archive.
- Open the central workbook and verify the sheets Problem Catalog, Material Solution Units, Cost Table, Role Execution Map, Physical Inventory, Pathway Library, Real Resolution Score, Complete Cases and Dashboard.
- Register date, operator, local context and package version in use.
- Do not modify formulas, thresholds, field names or canonical columns without a change log.
- Always use working copies for real cases; the original package remains the master.

15.2 Standard procedure from intake to solution

Step 1 - Intake. Create a problem_id and record domain, trigger, severity, urgency, hypothesized cause and data source. If data are incomplete, mark the case as LOW_CONFIDENCE and collect additional data before any high-risk decision.

Step 2 - Cataloging. Search the problem_code in the Problem Catalog. If the problem exists, apply the target factory and primary pathway. If it does not exist, create a provisional entry and classify it as PROBLEM_NEW_PENDING_REVIEW.

Step 3 - Material Solution Unit. Complete an MSU with material component, digital component, authorized actor, estimated cost, access pathway, funding pathway, outcome and fallback. No problem should remain without an MSU or a reasoned declaration of impossibility.

Step 4 - Costs. Consult the Cost Table and distinguish possible public cost, possible private cost, locally validated cost and fallback cost. If cost blocks the solution, activate the Funding Engine and the Public Responsibility Dossier.

Step 5 - Roles. Consult the Role Execution Map. Each action must have a role, maximum response time, required document and fallback. Unauthorized actors cannot perform clinical, experimental, biological or reserved legal functions.

Step 6 - Pathway. Select the pathway in the Pathway Library. Execute the steps in the defined order. If a step fails, record the reason and activate fallback 1; if fallback 1 fails, activate fallback 2 or a Public Responsibility Case.

Step 7 - Real Resolution Score. Calculate RRS before the intervention and after the first outcome. A low RRS does not mean the case is useless; it shows where material resolution was interrupted: unclear cause, missing solution, cost barrier, missing actor or no outcome improvement.

Step 8 - Outcome. Record outcome at 48 hours, 7 days, 30 days and 90 days where applicable. No case is closed without outcome, final status, fallback used and residual responsibility.

15.3 Non-modifiable operating rules

- The package does not execute clinical, biological, genetic, cellular or experimental interventions.

- Every activity reserved to clinicians, laboratories, public bodies or authorities is transformed into handoff, referral, dossier or formal request.
- If a solution exists but is inaccessible due to cost, the case is not closed: it moves to the Funding Engine and Public Responsibility Dossier.
- If a solution does not exist, the case becomes an Invention Route or EDEN/Lab Handoff toward authorized actors.
- If the problem is economic, housing-related, food-related or related to essential medication, the economic factory has operational priority.
- If pain is severe, the Pain Pathway must be activated before any happiness or longevity assessment.
- If a person is below a critical threshold, the system must not stop at measurement: it must produce a material route or record a public responsibility.

15.4 Rules for the three demonstrative cases

Severe pain case. Trigger: NRS greater than or equal to 7. Minimum output: pain pathway within 48 hours, authorized role, estimated cost, clinical access, NRS or functional outcome. If the public route does not respond, the case becomes a Public Responsibility Case.

Economic collapse case. Trigger: EFS equal to or below 14, or housing, food, medication or care below critical threshold. Minimum output: economic rescue, funding route, food/housing/medication/care pathway and EFS outcome. If the competent body does not respond, record Non-Response Accountability.

Inaccessible advanced therapy case. Trigger: existence of an authorized therapy or potentially relevant trial, but access blocked by cost, eligibility, geography or documents. Minimum output: eligibility check, access map, authorized center, funding route, lab/clinical handoff and public fallback.

15.5 Minimum completion criteria

- The problem has a code or provisional entry.
- The MSU is complete.
- The pathway has been selected.
- Costs have been estimated or marked as locally to be validated.
- The responsible role is identified.
- Fallbacks are defined.
- RRS has been calculated or declared non-calculable with reason.
- The outcome has been recorded or scheduled.
- Every economic, public, clinical or laboratory block has a corresponding dossier.

16. Operational appendices

Appendix A - Minimal Problem Catalog

The Problem Catalog is the technical entry point of the system. Every problem must contain problem_id, problem_code, domain, trigger, severity, urgency, root cause, factory target, MSU_required, primary pathway, fallback 1, fallback 2 and required output.

- P001 PAIN_SEVERE: NRS greater than or equal to 7; Physical Care Factory; pathway PATH_PAIN_48H.
- P002 MEDICATION_MISSING: essential medication unavailable; Physical Care and Economic Security; pathway PATH_MEDICATION_ACCESS.
- P003 HOUSING_CRITICAL: EFS_2 less than or equal to 1; Economic Security Factory; pathway PATH_HOUSING_RESCUE.
- P004 FOOD_CRITICAL: EFS_3 less than or equal to 1; Economic Security Factory; pathway PATH_FOOD_SECURITY.
- P005 CARE_ACCESS_BLOCKED: needed visit, care or exam not reached; Physical Care and Public Responsibility; pathway PATH_CARE_ACCESS.
- P006 ADVANCED_THERAPY_INACCESSIBLE: existing advanced therapy not reachable; Advanced Therapy Access Factory; pathway PATH_ADVANCED_ACCESS.
- P007 UNRESOLVED_BIOLOGICAL_GAP: biological problem without sufficient solution; EDEN/Lab Factory; pathway PATH_LAB_HANDOFF.
- P008 LEGAL_ACCESS_COLLAPSE: procedure or rule produces denied access; Legal and State Responsibility Factory; pathway PATH_LEGAL_COLLAPSE.

Appendix B - Minimal Cost Table

The Cost Table is not a universal price list. Every amount is a provisional range to be locally validated. The function of the table is to prevent cost from remaining invisible.

- Essential medication: local cost to be verified; public route possible if covered by public health system, exemption or fund.
- Basic or specialist medical visit: public cost possible if accessible through the health system; private cost variable.
- Ordinary diagnostic exam: public cost possible; private cost to be estimated for urgency or fallback.
- Food support: often low-cost or public through social services, vouchers or charity.
- Housing support: highly variable; requires local validation and public actor.
- Health transport or access to a center: often underestimated; must be recorded.
- Documents, scanning, printing and postage: low cost, but may block access in fragile cases.
- Biological age clock panel: advanced, non-essential, non-standard; use only as optional outcome.
- Advanced therapy: high or very high cost; activate funding route and public responsibility.

Appendix C - Minimal Role Execution Map

- Case coordinator: collects data, completes MSU, activates pathway, does not prescribe and does not decide therapies.
- Authorized clinician: assesses, diagnoses, prescribes or refers according to competence.
- Social worker or equivalent body: assesses housing, food, income, benefits and material supports.
- Authorized laboratory: performs only tests and validations allowed under local rules.

- University or research center: validates methods, protocols, hypotheses and data, without unauthorized clinical application.
- Foundation or funder: covers bridge costs or pilots with transparency and without predatory control.
- Public authority: receives responsibility dossiers, responds, funds or gives reasoned refusal.
- Founder steward: protects trademark, direction, integrity and authorized use of the system.

Appendix D - Minimal Physical Inventory

- Tier 0: computer, smartphone, scanner or scanning app, printer, secure cloud, workbook, document templates.
- Tier 1: blood pressure monitor, scale, optional pulse oximeter, ordinary clinical tests, validated questionnaires.
- Tier 2: genetic or epigenetic testing only when indicated and authorized, specialist centers, trial registries.
- Tier 3: AI/EDEN hypothesis environment, authorized laboratory, preclinical or clinical validation where allowed.
- Low-cost substitutes: phone instead of complex platform, mobile scan instead of scanner, public service instead of private, telemedicine where appropriate.

Appendix E - Minimal Pathway Library

- PATH_PAIN_48H: confirm pain, verify urgency, activate doctor or pain therapy, record NRS at 30 and 90 days.
- PATH_MEDICATION_ACCESS: identify medication, economic or prescription barrier, activate equivalent, exemption, fund or public request.
- PATH_ECONOMIC_RESCUE: calculate EFS, identify housing/food/medication/care, activate material resource and funding route.
- PATH_MENTAL_HEALTH: assess PHQ-9/GAD-7 and acute risk, activate competent professional or service, reduce material causes.
- PATH_LONGEVITY_L0: use ordinary biomarkers, function, sleep, pain, movement, nutrition and follow-up without unvalidated anti-aging claim.
- PATH_ADVANCED_ACCESS: verify diagnosis, therapy, center, documents, cost, coverage, funding and public fallback.
- PATH_LAB_HANDOFF: transform unresolved problem into hypothesis, target, validation question and authorized partner.
- PATH_PUBLIC_DUTY: produce dossier, cost of inaction, concrete request, response deadline and non-response log.

Appendix F - Operational Real Resolution Score

The Real Resolution Score measures how far a problem has been transformed into material resolution. The seven components are: problem detected, cause clarified, solution available, physical deliverability, funding access, authorized execution and outcome improvement. Each component is scored from 0 to 1. The final score is the weighted mean in the workbook. If it cannot be calculated, specify which component is missing.

Appendix G - Cost of Inaction

The Cost of Inaction estimates the harm of non-intervention: clinical worsening, prolonged pain, economic loss, avoidable hospitalization, loss of housing/food/medication, isolation, delays and future public cost. Absolute precision is not required at the initial stage; the purpose is to make inertia visible.

Appendix H - Public and funding dossiers

Each dossier must contain problem, available evidence, requested solution, estimated cost, cost of inaction, responsible actor, concrete request, response deadline, expected outcome and fallback. The tone must be technical, measurable and non-ideological.

Appendix I - Authorized laboratory handoff

The Lab Handoff does not contain operative experimental protocols. It contains problem, context, hypothesis, target, available data, validation question, evidence level, risks, authorization boundaries and outcome to be measured. Every experimental activity remains the exclusive competence of authorized actors.